

3.9 Exponential & Logarithmic Derivatives Worksheet

1. Given $g(t) = e^{t^3}$, find its second derivative $g''(t)$.
2. Compute $f'(r)$ given that $f(r) = \tan^{-1}(\sqrt{r^2 + 5})$.
3. Given $h(t) = \ln(t^3 + 4t^2 + 6)$, find its second derivative $h''(t)$.
4. Compute $h'(t)$ if $h(t) = \ln(\tan(t^5 + 2t))$.
5. Compute $f'(x)$ if $f(x) = 4^x \cdot \sin(6x)$.

6. Compute $\frac{dy}{dx}$ given each of the following equations.

a) $y = \sqrt{\arcsin(e^{-5x})}$

b) $y = (x^2 + 9)^{3\ln(x)}$

c) $f(x) = \ln\left(\sqrt[3]{\sin(x^5 + 4x)}\right)$

d) $y = \frac{(3x^2 - 1)^2}{\sin(e^x)}$

e) $g(t) = \sin^{-1}(e + 2t)$

f) $h(x) = \log_4(7^{\pi x})$

g) $y = [\sin(x)]^{5x^2}$

h) $y = [\ln(x)]^{\cos(x)}$

i) $\ln(y^3) = x^3 + y^5$

7. Use logarithmic differentiation to find the derivative of $y = \frac{x^3 \sqrt{x^4 + 3}}{(5x + 1)^8}$.

8. Find $g'(x)$ if $g(x) = \cos^{-1}(e^{2x})$.